

RESOLVE & SUPPORT, REIMAGINED FOR HUTCH

Hutch Clarity

Explain every rupee. Fix it by rule. Prove it won't happen again.

One platform for customers and staff — on the website, the Hutch app, WhatsApp and the staff console.

Copilot

Trust Receipt

Autopsy

Foresight

Clarity Desk



THE PROBLEM

Money moves. Nobody can say why.



“The bank took 3,500 for my 90-day pack twice. My Hutch balance is still zero.”

Reload taken twice · Reddit r/srilanka, Aug 2024



“I recharged and didn’t buy anything — the money was deducted anyway.”

Surprise VAS debit · Reddit r/srilanka, Dec 2025



“The app says 4 days left. My ‘unlimited’ data just stopped.”

Hidden fair-use cap · Reddit r/srilanka, 2025–26



“WhatsApp four times, plus an email. No reply after a week.”

Support can’t resolve · Trustpilot, Aug 2025

What customers have today

- ✓ Balance by SMS, USSD and app
- ✓ Daily totals in the Hutch app
- ✓ *5510#: last 5 events, data only as of midnight
- ✓ Complaints via 1788, WhatsApp, email, app form

What is missing

- ⚠ The exact reason for a deduction
- ⚠ One case file across payment, charging, pack and consent logs
- ⚠ Safe automatic fixes for known causes
- ⚠ A receipt after a correction
- ⚠ The same answer on every channel, every time
- ⚠ Snowflake insight (12 h → 6 h) never reaches the chat

From insight to one platform

Customers don't hate rules. They hate surprises they can't check.



The answer is in the logs

Split across payment, charging, catalogue, consent and usage.



Acting beats advising

Hostinger: ~20% → 85–91% resolved once its agent could act.



The rulebook exists

Gazette 2316/14: VAS needs consent + OTP; logs producible.



Copilot

Finds the exact cause of any charge and fixes safe cases in one tap.



Trust Receipt

Signed, QR-checkable proof of what happened, what we fixed, what protects you.



Autopsy

Turns thousands of messy complaints into root causes and self-service flows.



Foresight

MiroFish-style swarm simulation predicts complaints before a launch.



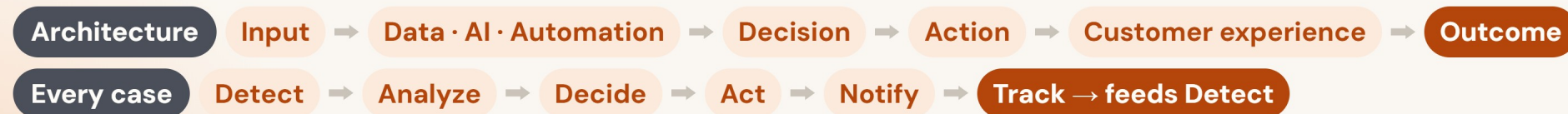
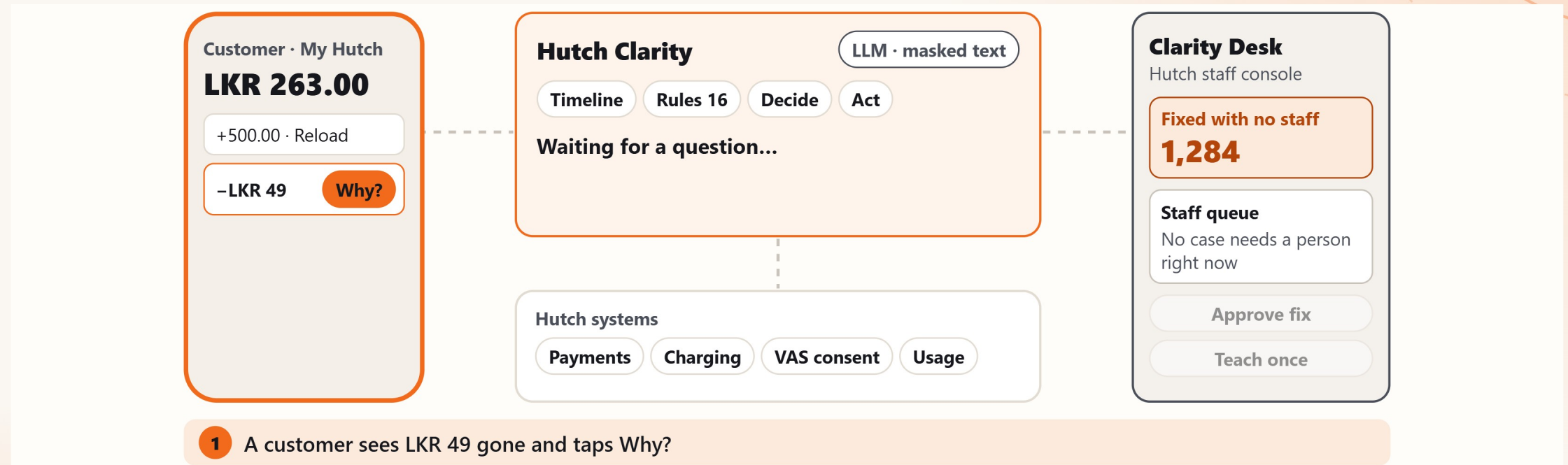
Clarity Desk

Staff console: every conflict on one screen; fix one case or ten thousand.

Foresight predicts → Autopsy spots the pattern → Copilot fixes → Desk handles the hard ones → Receipts prove it

SEE IT IN ACTION

How customers and staff use Clarity



FOR CUSTOMERS

One tap to a receipt, and help every day

- 1 Tap Why? or say it**
Web, app or WhatsApp;
Sinhala, Tamil,
English, Singlish
- 2 Get the reason**
With evidence, and what
was ruled out
- 3 Confirm in one tap**
Charge reversed, safeguard
switched on
- 4 Keep the receipt**
Signed, shareable, same
record staff see

 **Zero-contact:** a double reload is spotted in the live event stream and refunded before the customer even notices.

-  **Pack truth label**
Fair-use cap, after-cap speed and
apps shown before buying
-  **Talk, don't type**
Voice note in Sinhala or Tamil in,
spoken answer back
-  **Family guardian**
Look after up to 10 family numbers
and their safeguards
-  **Why? on any phone**
No smartphone? A short code sends
the reason by SMS
-  **Choose before you pay**
Pack ending? Stop data, cap spend or
top up — no silent burn
-  **One case, any channel**
Start on WhatsApp, finish in the app
or at a shop

Proactive care, and proof they keep

Bill-shock risk: 78% · High

Why? Pack 92% used, 2 days left · data stays on after the pack ends · LKR 480 burned the same way last month

Recommended action: ask now, in Sinhala — stop data, cap at LKR 100, or top up

PROACTIVE

- ✓ Refunds before they ask
- ✓ Pack-end choice, early
- ✓ VAS renewal, 1-tap stop
- ✓ Fair-use alerts 80% / 95%
- ✓ Outage heads-up + ETA

PERSONAL

- ✓ Language remembered
- ✓ Short or long answers
- ✓ Safeguards per person
- ✓ Family guardian mode
- ✓ Voice and large text

TRUST RECEIPT · TR-2026-000184

Verified

WHAT HAPPENED

A daily game subscription (LKR 49.00) was charged at 14:06 with no OTP from you.

WHAT WE CORRECTED

+ LKR 49.00 returned

Subscription off · Balance 263.00 → 312.00

Safeguard: merchant blocked until you opt in

Test PASSED



Scan to verify. Quote TR-2026-000184 on 1788, WhatsApp or to TRCSL.

WHY PEOPLE CAN TRUST IT

- 🔒 Ed25519-signed, hash-chained · QR check page
- ✅ Recurrence test PASSED only if really blocked
- 🌐 si · ta · en · PNG, PDF, SMS · one record for all

Rules decide. The LLM explains.



Grey = LLM, language only · Orange = deterministic rules that decide and move money

VAS_NO_CONSENT · confidence 96%

Why? LKR 49 at 14:06 · no OTP or second confirmation · Gazette 2316/14

Ruled out: pack expiry (3.2 GB left) · loan recovery (none in 30 days)

Action: refund LKR 49 · switch off · block merchant until opt-in

GUARDRAILS

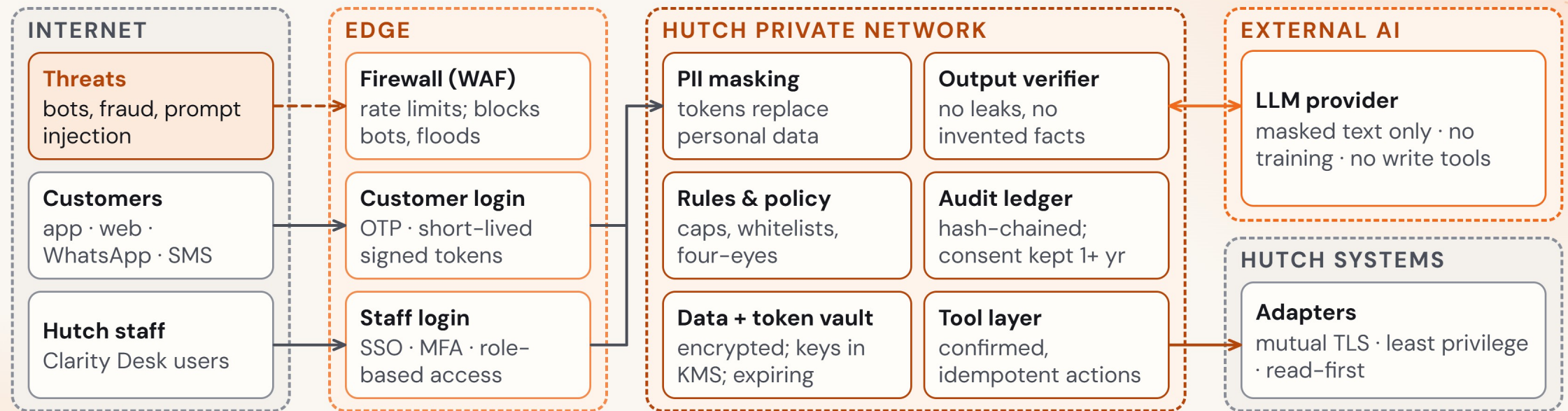
- 🔒 Text is a hint, never evidence
- 🔍 Missing log → a human, never a guess
- ✓ Works without the LLM (templates)

Outcome	When it applies	Example
Auto-fix	Money back only · high confidence · under the cap	Double reload refunded unasked
Fix with one tap	A service changes too · the customer confirms	VAS without OTP; wrong pack (even if used)
Staff approval	Above the cap, recent SIM swap, or two causes close	Large disputed reload
Explain only	Rule the customer saw: explained, not refunded	“Unlimited” hit a FUP shown at purchase
Hand-off	Log missing, fraud risk, or the customer asks	Staff start with the full trail

Private and secure by design

PII masking → Detect → Mask → tokens → Send only facts → Check reply → Restore inside Hutch

Sent: number 078 123 4567, NIC 199512345678, Nimal → LLM sees: number <PHONE_1>, NIC <NIC_1>, <NAME_1>



 **Watch everything:** SIEM alerts · refund-anomaly detection · TRCSL audit exports · penetration tests · **Privacy:** OTPs and cards never sent · masked logs and traces · tokens expire with the case · PDPA No. 9 of 2022 · Foresight on aggregates only

Every conflict on one screen



Queue · case timeline with evidence · ranked causes · one-click fix · reply box · teach bar

EVERY DAY

Case cockpit — evidence, ranked causes and what was ruled out

One-click decisions — the recommended fix, already policy-checked

Reply in their language — read in English, send in Sinhala or Tamil

Smart queue — sorted by money at stake, repeats and frustration

Teach once — a correction becomes a test case and a rule proposal

BIG AND TRICKY CONFLICTS



Fix all like this

One fix for thousands,
each receipted



Second look

Fresh reviewer: their
words vs logs



Policy what-if

Replay past cases
before a rule change



Merchant watch

Score VAS merchants;
suspend bad ones



Shift handover

Auto summary: cases,
promises, owners



Regulator pack

One click: trail +
consents for TRCSL

Know where to improve, then improve it



1 · Top services

Most-asked services by volume, trend and cost

2 · Where the AI stops

Every hand-off records why: counted, not guessed

3 · Self-service drops

Funnels show the exact step people quit

4 · Slice it any way

Language, channel, region, segment, plan type

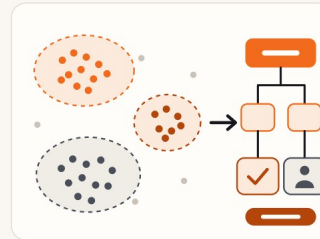


AI can't handle yet	Why	What we do	Owner
Log missing for a charge	A system lacks a field	Add the field; staff checklist until then	IT / VAS team
A brand-new problem	No rule exists yet	Teach-once cases draft a rule, then test it	CX engineers
Low confidence in Tamil	Too few local examples	Add real phrases to tests and prompts	AI team
Asked for a person early	Answer felt generic	Offer a person sooner; shorter answers	CX design
Amount above the cap	Large disputed charge	One-click approval; caps tuned by what-if	Finance
Network or coverage fault	Needs field work	Auto ticket with cell, time and honest ETA	Network ops
Self-service drop-off	One confusing step	Rewrite the step, test two versions	CX design
Product fault found	Sold pack ≠ given pack	Fix provisioning check; alert sales	Product team

Learn from complaints, predict the next

Complaint Autopsy

Fix a cause once, for everyone

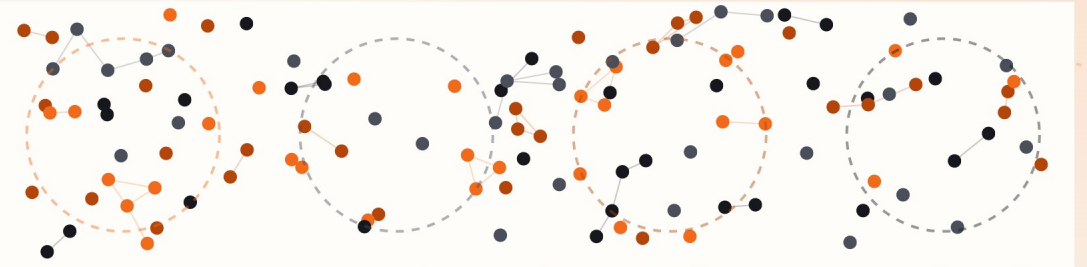


- 1 **Clean & protect** — dedupe, detect language, mask PII first
- 2 **Understand** — LLM canonical summary, so Singlish clusters
- 3 **Cluster** — multilingual embeddings + UMAP + HDBSCAN
- 4 **Generate the flow** — self-service from real checks only
- 5 **Test & publish** — replayed, approved, live on WhatsApp

CAUSES IT ALREADY KNOWS

VAS surprise Reload issue Pack mismatch Social pack
FUP stop Balance burn Pack sunset No reply

Foresight · rehearse tomorrow today



- 1 **Seed the change** — new pack, price, policy or outage
- 2 **Build the world** — students, dual-SIM, tourists, parents
- 3 **Simulate** — thousands of agents react, round after round
- 4 **Act early** — predictions become flows, scripts, fixes

Example: retiring a popular pack → predicted complaints by segment, migration cards ready before launch.

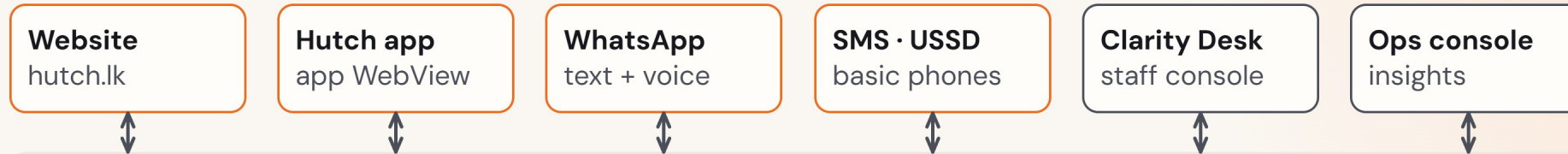
 **Early-warning radar:** live spikes alert staff before the queue fills.

SYSTEM ARCHITECTURE

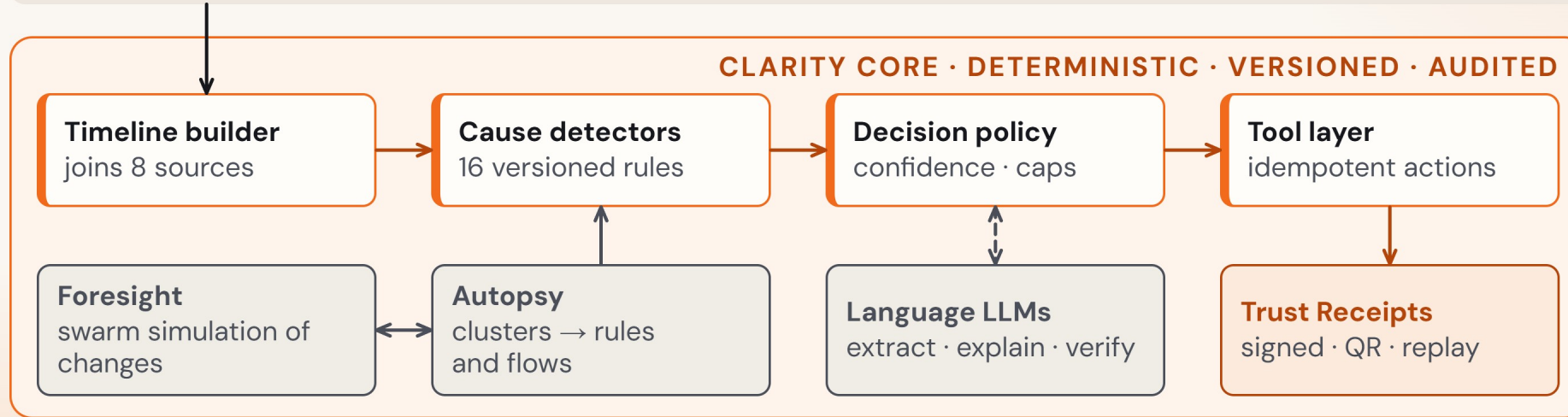
How Hutch Clarity is built

Orange = deterministic rules that move money · Grey = AI for language, patterns and prediction · Deep orange = proof for the customer

CHANNELS



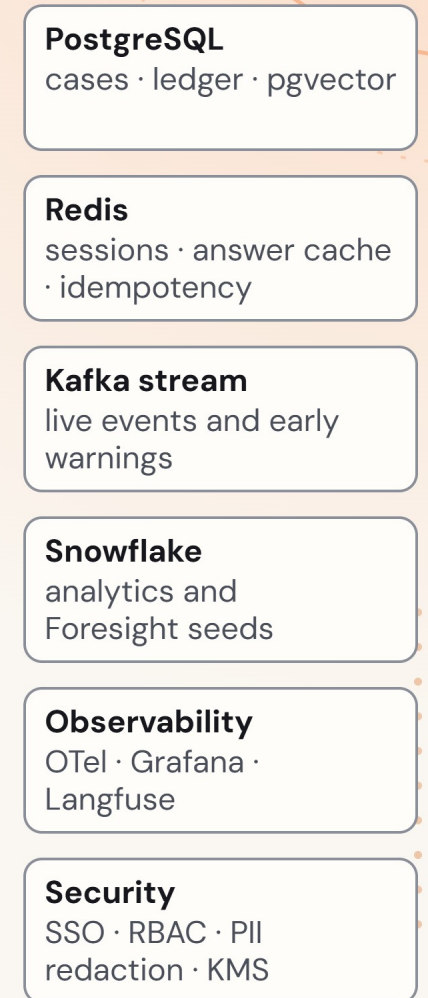
ORCHESTRATION · identity (SSO / OTP) · language detection (Sinhala · Tamil · English · Singlish) · message batching · handoff check on every turn · router to disputes, policy answers or self-service flows



ADAPTERS · TM Forum API shapes · read into the timeline · write only via the tool layer



DATA & PLATFORM



TECHNOLOGY

Every technology has a job

Area	Our choice	Its job
Generative AI / LLMs	Reasoning + non-reasoning LLMs · TTS	Explain in si/ta/en; voice replies
ML & prediction	Multilingual embeddings · UMAP · HDBSCAN	Clusters, bill-shock risk, confidence
NLP	LLM extraction + canonical summaries	Understands Singlish and Tamil
RAG & agentic AI	Catalogue, T&C, Gazette retrieval · MCP tools	Cited answers; tools need confirmation
Swarm prediction	MiroFish engine (OASIS · GraphRAG · Zep)	Rehearse changes before launch
APIs & microservices	Python 3.12 · FastAPI · Pydantic · Celery	Same core stack as Hostinger's agent
Automation & rules	Versioned rule packs · OPA · stream detectors	Deterministic, tested, auditable
Data & visualization	PostgreSQL 16 + pgvector · Redis · Kafka	Money records, cache, live events, KPIs
Web & mobile	Next.js · TypeScript · Tailwind · PWA	One codebase: web, app, consoles
Conversational AI	WhatsApp Cloud API · SMS / USSD gateway	Where customers already are
Receipts	Playwright render · Ed25519 · QR	Correct Sinhala/Tamil; tamper-evident
Integration	Adapters on TM Forum Open APIs	Plugs into Hutch systems by config
Cloud & delivery	Docker · Kubernetes · CI · OpenTelemetry	Autoscaling; every call traced

Safe to automate, cheap to run

Engineering value

-  **Rules own the money**
Deterministic, versioned rules; any decision replays
-  **AI kept in its lane**
LLMs only read and write; verifier + templates
-  **Safe by construction**
Idempotency, caps, refund budget, reconciliation
-  **Plug in, don't rip out**
TM Forum adapters, shadow mode first, no rewrite
-  **Build once, reuse**
One timeline, flow definition and frontend for all
-  **Improves every week**
Traces, golden tests, Teach once feed new rules
-  **Real-time and scalable**
Kafka events; scales out by phone number

Answer without the LLM when we can

- 1 Rules + templates** **0 tokens**
Disputes: trail facts, approved wording
- 2 Redis answer cache** **0 tokens**
Common questions, exact or by meaning
- 3 Non-reasoning LLM** **Few tokens**
Short, simple new questions
- 4 Reasoning LLM** **Most tokens**
Complex or unusual cases only

HOW THE CACHE STAYS SAFE

- Generic answers only** Keyed by language + catalogue
- CX-approved answers** Masked text; entries expire
- Hit rate shown in Desk**

Built-in qualities, greener by design

Real-time

Live events trigger fixes and warnings

Explainable AI

Cause, evidence, ruled-out options

Personalized

Own language, channel, safeguards

Proactive

We reach out first, before a complaint

Automated

Safe causes fixed; one fix for thousands

Accessible

Voice, USSD/SMS, 3 languages, large text

Green by design

Paperless, fewer trips, lean AI

Scalable

Stateless services, scale by number

API-driven

Every action a typed API (MCP too)

Reusable

One timeline, flow DSL and UI kit

Secure

SSO, RBAC, PII masking, signed receipts

Measurable

KPIs from day one, proven in pilot

Greener support

Paperless proof

Digital receipts replace paper and paperwork

Fewer trips and calls

First-contact fixes cut calls and shop visits

Right-sized AI

Rules and cache first; big LLM only if needed

Light on devices

Lean web module for older phones, little data

Efficient cloud

Autoscaling; Foresight in small batches

We will report model calls avoided, shop visits avoided and energy per case.

Value for customers and for Hutch

BEFORE · THE FRICTION

Money gone, no reason given
Days waiting on WhatsApp or 1788
Different answers every time
Surprise burn when a pack ends

DURING · HOW CLARITY HELPS

Tap Why? or send a voice note
Reason with evidence in seconds
One-tap fix; staff when needed
Receipt in Sinhala, Tamil, English

AFTER · WHAT IMPROVED

Answer and fix in minutes
Proof kept, same story everywhere
A safeguard stops a repeat
Fewer complaints every month



Keeps customers

Explained charges stop quiet churn



Protects VAS revenue

Consent proof; bad merchants out



Lowers cost to serve

Auto-fixes, trails and bulk fixes



Cuts regulatory risk

Gazette proof, TRCSL packs, PDPA



Fewer launch problems

Faults found before launch



A trust brand

The network that explains every rupee

IMPROVEMENTS WE WILL MEASURE

Shorter resolution time

Fewer complaints

Less bill shock

More self-service

Higher satisfaction

More engagement

Lower churn

Lighter staff workload

Better accessibility

HUTCH CLARITY · THE ROAD AHEAD

From “we don’t know” to “here is your receipt”.

Explain every rupee. Fix it by rule. Prove it won’t happen again — and see the next problem coming.

- Step 1 · **Shadow mode** — read-only on anonymized logs; suggests causes to staff
- Step 2 · **Clarity Desk live** — staff-approved fixes and receipts; weekly Autopsy
- Step 3 · **Customer launch** — Why? in app, web and WhatsApp; whitelisted auto-fix
- Step 4 · **Foresight & scale** — every launch rehearsed; postpaid, home broadband

Thank you · Questions welcome

